MATH 110 — Week 8

Notes Chapters 14–15; Homework 8

§1 Successive kernels and generalized eigenvectors

The question. What replaces eigenvectors when they do not span? For $N = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$, $Ne_2 = e_1 \neq 0$ but $N^2 e_2 = 0$: the second power detects the missing direction.

Tools from earlier weeks.

- Weeks 2–4: extending a kernel basis to a domain basis makes the remaining images a range basis; hence $\dim \ker A + \dim \operatorname{range} A = \dim V$. An operator in finite dimension is invertible iff its kernel is zero. Two subspaces with zero intersection and dimensions summing to $\dim V$ are complementary (Week 2's two-of-three criterion).
- Weeks 5–7: $p(T) = \sum a_j T^j$, with $T^0 = I$. Distributing finite sums gives $(pq)(T) = p(T)q(T)$ and commutation, so T preserves polynomial kernels and ranges. Division by m_T proves $q(T) = 0$ iff $m_T \mid q$. Its roots are exactly the eigenvalues: if $m_T = (x - \lambda)q$, minimality supplies $q(T)w \neq 0$, an eigenvector.
- Week 7: an eigenbasis exists iff the eigenspaces sum to V, equivalently their dimensions sum to $\dim V$; distinct eigenspaces sum directly by applying shifted factors to a relation. An upper-triangular matrix has its diagonal entries as eigenvalues, by back substitution. Every operator on a nonzero finite-dimensional complex space has an eigenvalue by the fundamental theorem of algebra (Week 6, with cited analytic inputs).

Work on finite-dimensional V over a field F; complex or real hypotheses are stated where needed. Write $E(\lambda, T) = \ker(T - \lambda I)$ and $n = \dim V$.

Definition (Notes Definition 37; LADR 8.8, 8.19)**.** For T on finite-dimensional V, define $G(\lambda, T) = \bigcup_{k \geq 0} \ker(T - \lambda I)^k$. Its nonzero vectors are generalized eigenvectors for λ: they satisfy $(T - \lambda I)^k v = 0$ for some positive integer k.

$$v \xrightarrow{T-\lambda I} (T-\lambda I)v \xrightarrow{T-\lambda I} \cdots \xrightarrow{T-\lambda I} w \neq 0 \xrightarrow{T-\lambda I} 0$$

$$Tw = \lambda w$$

The last nonzero vector is an ordinary eigenvector: the next application gives $(T - \lambda I)w = 0$. In the question, $e_2 \mapsto e_1 \mapsto 0$.

Proposition (Notes Proposition 69; LADR 8.1–8.2)

For any operator A, $\{0\} = \ker A^0 \subseteq \ker A \subseteq \ker A^2 \subseteq \cdots$. If $\ker A^m = \ker A^{m+1}$, then every later kernel equals $\ker A^m$.

Why one equality is enough

Obstacle. The equality $\ker A^m = \ker A^{m+1}$ only tells us: if $A^{m+1}u = 0$, then $A^m u = 0$. A vector satisfying $A^{m+2}v = 0$ has one power too many to use it directly.

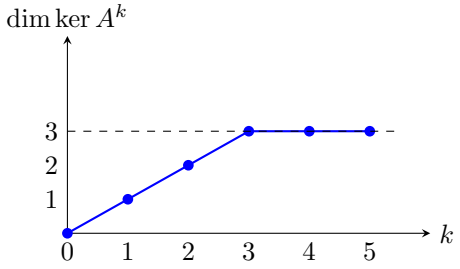
Choice. Set $u = Av$, since $A^{m+1}(Av) = 0$. The equality now removes one power. Apply it once more to v to remove the next. The same reduction works however far out we start.

Proof. 1. If $A^k v = 0$, then $A^{k+1} v = A0 = 0$, proving each inclusion.
2. If $A^{m+2} v = 0$, then $Av \in \ker A^{m+1} = \ker A^m$, hence $A^{m+1} v = 0$ and $v \in \ker A^m$.
3. Thus $\ker A^{m+2} = \ker A^m$; repeat the same argument inductively for every later power.

□

Proposition (Notes Proposition 70; LADR 8.3, 8.20)

If $n = \dim V$, then $G(\lambda, T) = \ker(T - \lambda I)^n$, an invariant subspace. The kernels of any operator A stabilize by exponent n.



$Ae_3 = e_2, \quad Ae_2 = e_1, \quad Ae_1 = 0$

Each strict inclusion adds a basis direction. The first equality prevents all later growth.

On F^3 the dimensions are $0, 1, 2, 3, 3, \ldots$. In dimension n, more than n strict increases from zero are impossible.

Proof. 1. If $\ker A^n \neq \ker A^{n+1}$, Proposition 69 implies every inclusion from $\ker A^0$ to $\ker A^{n+1}$ is strict. The last kernel would have dimension at least $n + 1$, impossible.
2. Thus $\ker A^n = \ker A^{n+1}$, and Proposition 69 gives equality with every later kernel. Earlier kernels are contained in it.
3. Use $A = T - \lambda I$ to identify the union. It is invariant because T commutes with $(T - \lambda I)^n$ (Week 6). The zero-dimensional case gives $\ker I = V = \{0\}$.

□

Proposition (LADR 8.4: stabilized kernel and range)

For $n = \dim V$ and $A \in L(V)$, $V = \ker A^n \oplus \operatorname{range} A^n$. Both summands are invariant under every operator commuting with A; A is invertible on $\operatorname{range} A^n$.

Why use a high power for both subspaces?

First try. Rank–nullity gives $\dim \ker A + \dim \operatorname{range} A = n$, but does not give zero intersection: for $Ae_2 = e_1$, $Ae_1 = 0$, the vector e_1 belongs to both.

Choice. Replace A by A^n, after the kernels have stopped growing. An intersection vector has the form $w = A^n v$ and satisfies $A^n w = 0$. Thus $A^{2n} v = 0$. Stabilization says that v was already in $\ker A^n$, making its image w zero. This supplies the missing zero-intersection condition.

Proof. 1. If $w = A^n v$ and $A^n w = 0$, then $A^{2n} v = 0$. Stabilization gives $A^n v = 0$, hence $w = 0$.
2. Rank-nullity gives $\dim \ker A^n + \dim \operatorname{range} A^n = n$. The zero intersection therefore makes their sum all of V, by Week 2's two-of-three criterion.
3. If B commutes with A, then $A^n Bv = BA^n v$ preserves the kernel, and $BA^n v = A^n Bv$ preserves the range. On $\operatorname{range} A^n$, a vector with $Aw = 0$ lies also in $\ker A^n$ (for $n > 0$), so is zero by step 1. The restricted operator is injective, hence invertible. For $n = 0$ it is the unique operator on the zero space.

□

Example (Notes Example 61)

For $A = \left(\begin{smallmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 3 \end{smallmatrix}\right)$ on $\mathbb{Q}^3$:

1. $A - I = \left(\begin{smallmatrix} 0 & 1 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 2 \end{smallmatrix}\right)$ sends (a, b, c) to $(b + c, c, 2c)$. Zero requires $c = b = 0$, so its kernel is $\mathbb{Q}e_1$.
2. $(A - I)^2 = \left(\begin{smallmatrix} 0 & 0 & 3 \\ 0 & 0 & 2 \\ 0 & 0 & 4 \end{smallmatrix}\right)$ and $(A - I)^3 = \left(\begin{smallmatrix} 0 & 0 & 6 \\ 0 & 0 & 4 \\ 0 & 0 & 8 \end{smallmatrix}\right)$ send (a, b, c) to $(3c, 2c, 4c)$ and $(6c, 4c, 8c)$. Each is zero iff $c = 0$, giving the same kernel $\operatorname{span}(e_1, e_2)$.
3. Proposition 69 gives $G(1, A) = \operatorname{span}(e_1, e_2)$; e_2 is generalized but not an ordinary eigenvector.

Example (LADR 8.6, 8.10, 8.21) 1. For $T(x_1, x_2, x_3) = (4x_2, 0, 5x_3)$ over $\mathbb{R}$ or $\mathbb{C}$, $\ker T = Fe_1$ and $\operatorname{range} T = \operatorname{span}(e_1, e_3)$, so their intersection is nonzero.

2. $T^3(x) = (0, 0, 125x_3)$ gives $\ker T^3 = \operatorname{span}(e_1, e_2)$ and $\operatorname{range} T^3 = Fe_3$, now complementary.
3. $(T - 5I)^3 x = (-125x_1 + 300x_2, -125x_2, 0)$ is zero iff $x_2 = x_1 = 0$. Thus $G(0, T) = \operatorname{span}(e_1, e_2)$ and $G(5, T) = Fe_3$. Ordinary eigenspaces are Fe_1 and Fe_3, so they do not span; the generalized eigenspaces do.

Proposition (Notes Proposition 71)

$G(\lambda, T) \neq \{0\}$ exactly when λ is an eigenvalue.

Proof. 1. An eigenvector satisfies $(T - \lambda I)v = 0$, so belongs to $G(\lambda, T)$.

2. Conversely, start with $v \in G(\lambda, T) \setminus \{0\}$ and apply $T - \lambda I$ until the last nonzero vector w (as in the figure). The next vector is zero, so $(T - \lambda I)w = 0$, or $Tw = \lambda w$. Thus λ is an eigenvalue.

□

Proposition (Notes Proposition 72; LADR 8.11–8.12)

Nonzero generalized eigenvectors for distinct eigenvalues are independent. In particular a generalized eigenvector corresponds to a unique eigenvalue, and the sum of distinct generalized eigenspaces is direct.

How to retain one contribution in a relation

Two eigenvalues. Suppose $av + bz = 0$, with $v \in G(\lambda, T)$ and $z \in G(\mu, T)$, $\lambda \neq \mu$. Applying $(T - \mu I)^n$ sends z to zero. We still need to know that it does *not* send v to zero.

Choice. First turn v into its last nonzero iterate $w = (T - \lambda I)^h v$. Then $Tw = \lambda w$, so $(T - \mu I)^n w = (\lambda - \mu)^n w \neq 0$. Apply the two operations to the whole relation; their commutation preserves the zero result on z. We get $a(\lambda - \mu)^n w = 0$, forcing $a = 0$. For several eigenvalues, use one such factor for each unwanted contribution.

Proof. 1. Suppose $\sum_{j=1}^{r} a_j v_j = 0$ with $v_j \in G(\lambda_j, T) \setminus \{0\}$. Fix an index i and choose the largest h for which $w = (T - \lambda_i I)^h v_i \neq 0$; it exists by the defining eventual zero property.

2. Then $Tw = \lambda_i w$. For $j \neq i$, $(T - \lambda_j I)^n w = (\lambda_i - \lambda_j)^n w$, with $n = \dim V$.
3. Apply $(T - \lambda_i I)^h \prod_{j \neq i} (T - \lambda_j I)^n$ to the relation. Commutation and Proposition 70 make every term except i zero.
4. The remaining equation is $a_i \prod_{j \neq i} (\lambda_i - \lambda_j)^n w = 0$. Each scalar factor and w is nonzero, so $a_i = 0$. Do this for every i.
5. If a nonzero v corresponded to two distinct eigenvalues, the dependent pair v, v would contradict independence. Grouping any relation with one vector from each generalized eigenspace proves directness.

□

Shortcuts

- Stop taking powers as soon as two consecutive kernels agree, by Proposition 69.
- Use exponent $\dim V$ as a guaranteed stopping point, by Proposition 70.
- A generalized eigenvector can be checked by a power equation even when it is not an ordinary eigenvector.

Traps

- A union of arbitrary subspaces need not be a subspace: the union of the coordinate axes omits $(1, 1)$. The nested-kernel union is a subspace because it equals one stabilized kernel.
- $\ker A$ and $\operatorname{range} A$ need not be complementary: the two-by-two N in the question has both equal to Fe_1. Use stabilized powers.

§2 Nilpotent operators and chains

The question. How can one choose coordinates for an operator whose power is zero? For differentiation on $\mathcal{P}_2(\mathbb{R})$, the ordered list $1, x, x^2/2$ has images $0, 1, x$, giving ones above the diagonal.

Definition (Notes Definition 39; LADR 8.14)**.** An operator N is nilpotent if $N^r = 0$ for some positive integer r, equivalently $G(0, N) = V$ by stabilization. On a nonzero finite-dimensional space its nilpotency index is the smallest such r. A matrix is strictly upper triangular if it is upper triangular with every diagonal entry zero. A chain of length r is an independent ordered list $v_1, \dots, v_r$ satisfying $Nv_1 = 0$ and $Nv_j = v_{j-1}$ for $j \geq 2$.

Proposition (LADR 8.16–8.18: nilpotence tests)

For $n = \dim V > 0$, a nilpotent N satisfies $N^n = 0$ and has only eigenvalue 0. It has minimal polynomial x^r, where r is its index. Equivalently, N has a strictly upper-triangular matrix in some basis. Over $\mathbb{C}$, having only eigenvalue 0 also implies nilpotence.

Choose the order of the basis from the powers

Matrix requirement. A strictly upper-triangular matrix means that the image of each basis vector uses only vectors listed earlier. How can we ensure this before computing any columns?

Choice. If $N^k v = 0$, then $N^{k-1}(Nv) = 0$. Hence N sends $\ker N^k$ into the preceding kernel. Start with a basis of $\ker N$, then add the vectors needed for $\ker N^2$, then for $\ker N^3$, and so on. Every added vector has its image in a space whose basis was already listed.

Proof. 1. If $N^r = 0$, then $\ker N^r = V$; stabilization implies $\ker N^n = V$, hence $N^n = 0$. If $Nv = \lambda v$ with $v \neq 0$, $N^r v = \lambda^r v = 0$ forces $\lambda = 0$; a nonzero nilpotent space has nonzero kernel, so 0 occurs.

2. Since $m_N \mid x^r$, it is a monic power x^s. Minimality of the index gives $s = r$. Conversely $m_N = x^r$ immediately gives nilpotence.
3. Extend a basis successively through $K_0 = 0 \subseteq K_1 = \ker N \subseteq K_2 = \ker N^2 \subseteq \cdots \subseteq K_n = V$. Since $N(K_k) \subseteq K_{k-1}$, every added basis vector maps into the span of earlier vectors. These image columns give a strictly upper-triangular matrix.
4. Conversely, a strictly upper-triangular matrix maps the first j basis vectors into the first $j - 1$ span; n applications give zero.
5. Over $\mathbb{C}$, if the only eigenvalue is 0, complex factorization and the root characterization of m_N (Week 7) give $m_N = x^r$. On the zero space use $m_N = 1$, the empty basis, and $N = 0$.

□

Example (LADR 8.15) 1. On F^4, $N(x_1, x_2, x_3, x_4) = (0, 0, x_1, x_2)$ has $N^2 = 0$, so is nilpotent of index 2.

2. Over $\mathbb{R}$ or $\mathbb{C}$, $B = \begin{pmatrix} -3 & 9 & 0 \\ -7 & 9 & 6 \\ 4 & 0 & -6 \end{pmatrix}$ satisfies $B^2 = \begin{pmatrix} -54 & 54 & 54 \\ -18 & 18 & 18 \\ -36 & 36 & 36 \end{pmatrix} \neq 0$ and $B^3 = 0$, so has index 3.
3. Differentiation on $\mathcal{P}_m(\mathbb{R})$ has index $m + 1$: the $(m + 1)$st derivative is zero and $D^m x^m = m! \neq 0$.

$$v \xrightarrow{N} Nv \xrightarrow{N} N^2v \neq 0 \xrightarrow{N} 0$$

coefficient a coefficient b coefficient c

After two applications, only the contribution starting at v can still be nonzero. This is why N^2 isolates a in the calculation below.

Proposition (Chain independence; LADR 8A Exercise 2)

If $N^r v = 0$ and $N^{r-1}v \neq 0$, then $N^{r-1}v, N^{r-2}v, \ldots, v$ is a chain of length r.

Find the argument on a three-vector chain

Small case. Suppose $N^3v = 0$ but $N^2v \neq 0$. We must show that a relation $av + bNv + cN^2v = 0$ has only zero coefficients. Applying N shifts every term one place to the right:

$$\begin{aligned} av + bNv + cN^2v &= 0 \quad \text{original relation,} \\ aNv + bN^2v &= 0 \quad \text{after applying } N, \\ aN^2v &= 0 \quad \text{after applying } N^2. \end{aligned}$$

Why it works. In the last equation $N^2v \neq 0$, so $a = 0$. The preceding equation then gives $b = 0$; the original gives $c = 0$. Applying a high power isolates a coefficient because the later iterates have already reached zero. The proof below does this for any length.

Proof. 1. Suppose $\sum_{j=0}^{r-1} a_j N^j v = 0$. Apply N^{r-1}. The $j = 0$ term becomes $a_0 N^{r-1}v$; every $j \geq 1$ term is zero because its exponent is at least r. Thus $a_0 = 0$.

2. After $a_0, \ldots, a_{h-1}$ have been shown zero, apply N^{r-1-h} to the remaining relation. It gives $a_h N^{r-1}v = 0$, since each $j > h$ again has exponent at least r. Hence $a_h = 0$. Induction gives all coefficients zero.

3. The list is independent. Its first vector maps to zero, and each following vector maps to its predecessor, by $N(N^j v) = N^{j+1}v$. It is therefore a chain.

□

Proposition (Notes Proposition 75; LADR 8.45)

Every nilpotent operator has a basis consisting of disjoint chains. Its matrix is zero except for possible ones immediately above the diagonal.

How to extend chains without losing their image equations

First try. One long chain need not fill V. Extending it to an arbitrary basis does not ensure that an added vector maps to an earlier one.

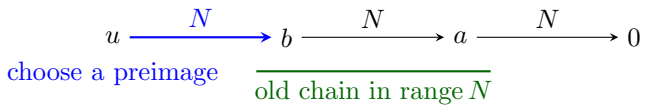
Choice. Work first in $U = \operatorname{range} N$, a smaller invariant space. Induction gives chains there. If a chain is $b \mapsto a \mapsto 0$, choose u with $Nu = b$; this is possible because $b \in \operatorname{range} N$. Now $u \mapsto b \mapsto a \mapsto 0$ is one step longer. Do this for every chain.

What is still missing? The lifted chains have every vector of U among their images. For any $v \in V$, choose a combination z of the lifted vectors with $Nz = Nv$. Then $v - z \in \ker N$. Thus only kernel directions can be missing; add those as one-vector chains.

Proof. 1. Induct on dimension. If $N = 0$, any basis consists of length-1 chains (including the empty basis when $V = 0$). Otherwise $U = \operatorname{range} N$ is proper: an invertible N could not have a zero power. Also $N(U) \subseteq U$, so $N|_U$ is nilpotent.

2. By induction choose a chain basis of U with top vectors $v_1, \dots, v_b$. Choose u_j with $Nu_j = v_j$. Adjoining u_j to its old chain extends that chain by one vector. Let L span all the extended chains; it contains U and $N(L) = U$.
3. These vectors are independent. Write a relation as $\sum_j c_j u_j + u = 0$, where $u \in U$ is expressed in the old basis. Applying N gives $\sum_j c_j v_j + Nu = 0$. The vector Nu uses only the old chain vectors *below* their tops, so independence of the old basis gives every $c_j = 0$. Then $u = 0$, and all its old-basis coefficients are zero.
4. The same calculation shows $L \cap \ker N = U \cap \ker N$, spanned by the bottom vector of each old chain. Extend those bottoms to a basis of $\ker N$, adding $w_1, \dots, w_\ell$. The extended chains and these new length-1 chains are independent: any relation would put a combination of the w_i in $L \cap \ker N$, contradicting the chosen kernel basis.
5. For $v \in V$, choose $z \in L$ with $Nz = Nv$, since $N(L) = U$. Then $v - z \in \ker N$. Thus $L + \ker N = V$, so the independent chains span V. In their bottom-to-top order, each nonzero image is the preceding basis vector, giving the asserted unit entries.

□



Lift the top b of each chain in range N. Adding a vector at the top preserves all old image equations. Missing vectors in $\ker N$ become separate length-1 chains.

Example (Notes Proposition 75; LADR 8.42–8.43) 1. On $\mathbb{C}^4$, $T(x) = (0, x_1, x_2, x_3)$ has chain

$$(T^3 e_1, T^2 e_1, T e_1, e_1) = (e_4, e_3, e_2, e_1).$$

Its matrix in that basis has three superdiagonal ones.

2. On $\mathbb{C}^6$, $T(x) = (0, x_1, x_2, 0, x_4, 0)$ has chains (e_3, e_2, e_1), (e_5, e_4) and (e_6). Their lengths are 3, 2 and 1. A single length-6 chain is impossible because $T^3 = 0$, so the first three entries in $T^5 v, \dots, v$ would be zero.
3. The two matrices displayed with Notes Proposition 75 are

$$\begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \qquad \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

Their chain lengths are $(2, 2)$ and (4), and their indices are 2 and 4. Applying the first matrix twice gives zero; applying the second three times still sends e_4 to $e_1 \neq 0$.

Shortcuts

- A longest chain has length equal to the nilpotency index, by its last nonzero iterate and the chain-basis theorem.
- Over $\mathbb{R}$ or $\mathbb{C}$, turn derivatives into unit chain steps using $1, x, x^2/2!, \dots, x^m/m!$.

Traps

- Over $\mathbb{R}$, only real eigenvalue 0 does not imply nilpotence: $\left(\begin{smallmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{smallmatrix}\right)$ has only real eigenvalue 0 but its square is nonzero and its rotation part never becomes zero.
- Nilpotent does not mean zero: $\left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$ is a nonzero example.

§3 Generalized eigenspace decomposition and characteristic polynomial

The question. Do generalized eigenspaces account for every vector? For $T(x) = (4x_2, 0, 5x_3)$, $\mathbb{C}^3 = G(0,T) \oplus G(5,T)$ even though ordinary eigenspaces do not span.

Proposition (Notes Proposition 73; LADR 8.9, 8.22)

For a finite-dimensional complex space and all distinct eigenvalues $\lambda_1, \dots, \lambda_r$, $V = \bigoplus_j G(\lambda_j, T)$. Each summand is invariant; on it $T = \lambda_j I + N_j$ with N_j nilpotent. In particular, V has a basis of generalized eigenvectors.

Why the remaining space is smaller

Choice. Apply the kernel–range split to $A = T - \lambda I$. It separates $G(\lambda, T)$ from $U = \operatorname{range}(T - \lambda I)^n$. The first space is nonzero because it contains an eigenvector, so $\dim U < n$.

Why induction applies. T preserves U, and $T - \lambda I$ is invertible there by the splitting result. Thus λ does not reappear on U. Over $\mathbb{C}$, any nonzero remaining space has an eigenvalue, so the same construction can continue until its dimension is zero. Over $\mathbb{R}$ a rotation may have no eigenvalue, so this starting step can fail.

Proof. 1. Induct on $n = \dim V$, with the empty basis at $n = 0$. For $n > 0$ choose an eigenvalue λ. Apply LADR 8.4 to $A = T - \lambda I$: $V = G(\lambda, T) \oplus U$, where $U = \operatorname{range}(T - \lambda I)^n$ is invariant.

2. The first summand is nonzero, so $\dim U < n$. Induction supplies a basis of U made of generalized eigenvectors of $T|_U$, also generalized eigenvectors of T.

3. Any basis of $G(\lambda, T)$ consists of generalized eigenvectors. Adjoining it gives a generalized eigenvector basis of V, proving that the sum of all $G(\lambda_j, T)$ spans. Proposition 72 makes the sum direct.

4. Each summand is invariant by Proposition 70. On it $(T - \lambda_j I)^n = 0$, so $N_j = (T - \lambda_j I)|_{G(\lambda_j, T)}$ is nilpotent, as asserted.

□

Definition (Notes Definition 38; LADR 8.23, 8.26)**.** The algebraic multiplicity of λ is $d_\lambda = \dim G(\lambda, T)$; its geometric multiplicity is $g_\lambda = \dim E(\lambda, T)$ (Week 7). Over $\mathbb{C}$, the characteristic polynomial is $\chi_T(x) = \prod_\lambda (x - \lambda)^{d_\lambda}$ over the distinct eigenvalues. The empty product for $V = 0$ is 1.

Proposition (Notes Definition 38, consequences; LADR 8.25, 8.28)

Over $\mathbb{C}$, $\sum_\lambda d_\lambda = \dim V$, $g_\lambda \le d_\lambda$, and T is diagonalizable exactly when $g_\lambda = d_\lambda$ for every eigenvalue. Also $\deg \chi_T = \dim V$, and its roots are exactly the eigenvalues.

Proof. 1. Proposition 73 and the dimension formula for a direct sum give $\sum d_\lambda = \dim V$. Containment $E(\lambda, T) \subseteq G(\lambda, T)$ gives $g_\lambda \le d_\lambda$.

2. By Week 7, an eigenbasis exists exactly when $\sum g_\lambda = \dim V = \sum d_\lambda$. Since each difference is nonnegative, this holds exactly when every difference is zero.

3. The defining product for χ_T has degree $\sum d_\lambda$ and roots precisely the listed eigenvalues, proving the remaining claims.

□

Proposition (Notes Proposition 74; LADR 8.29–8.30: Cayley–Hamilton)

Over $\mathbb{C}$, $\chi_T(T) = 0$. Consequently $m_T \mid \chi_T$; if $\deg m_T = \dim V$, then $m_T = \chi_T$.

Why. On the d_λ-dimensional space $G(\lambda, T)$, the shift is nilpotent, so its d_λth power is zero. This explains the exponent in χ_T: its matching factor sends that entire summand to zero, and commutation lets that factor act first.

Proof. 1. On $G(\lambda, T)$, the nilpotent N_λ satisfies $N_\lambda^{d_\lambda} = 0$ by LADR 8.16. Hence $(T - \lambda I)^{d_\lambda}$ is zero on that summand.
2. The commuting product defining $\chi_T(T)$ contains this factor, so is zero on each generalized eigenspace. Proposition 73 says their sum is all of V, so $\chi_T(T) = 0$.
3. Week 7's divisibility criterion now gives $m_T \mid \chi_T$. Both polynomials are monic, and $\deg \chi_T = \dim V$; equality of degrees makes their quotient 1. The case $V = 0$ gives $1(T) = I = 0$.

□

Proposition (LADR 8.31: multiplicities in triangular form)

In any upper-triangular matrix of a complex operator, the number of diagonal appearances of λ is d_λ. Thus χ_T is the product of the linear factors from its diagonal entries.

Why a bound on each multiplicity is enough

First try. Count the nonzero diagonal entries of $(T - \lambda I)^n$. This gives independent columns, but other columns may also contribute to the rank, so it gives only $d_\lambda \le b_\lambda$, where b_λ is the number of diagonal entries equal to λ.

Choice. Do this for every eigenvalue before trying to improve the bound. Generalized decomposition gives $\sum d_\lambda = n$, and the diagonal counts also sum to n. If even one d_λ were smaller, the total would be smaller. Thus each bound is equality. For diagonal counts $(2, 1)$, the bounds $d_1 \le 2$, $d_3 \le 1$ together with $d_1 + d_3 = 3$ force $(d_1, d_3) = (2, 1)$.

Proof. 1. In a triangular matrix with b zero diagonal entries, the columns with nonzero diagonal are independent: in a proposed relation, the greatest selected index with nonzero coefficient has a nonzero coordinate there. Thus rank is at least $n - b$, and nullity at most b.
2. Products of triangular matrices are triangular and their diagonal entries multiply, since $(AB)_{ii} = \sum_k a_{ik} b_{ki}$ has only $k = i$ nonzero. Therefore $(T - \lambda I)^n$ has as many zero diagonal entries as T has entries equal to λ.
3. Step 1 gives $d_\lambda = \dim \ker (T - \lambda I)^n \le b_\lambda$, its diagonal count. Every diagonal entry is an eigenvalue (Week 7), so $\sum b_\lambda = n = \sum d_\lambda$.
4. Each nonnegative difference $b_\lambda - d_\lambda$ must be zero. Substitute these multiplicities into the defining product for χ_T.

□

Connection to the earlier determinant formula (Notes Chapter 14 remark). Here det denotes the usual determinant from MATH 54. Citing its familiar rules (similar matrices have the same determinant; a triangular determinant is the product of its diagonal entries), choose a triangular matrix $B = S^{-1}AS$ for the complex operator. Then

$$\det(xI - A) = \det(xI - B) = \prod_{j=1}^{n} (x - b_{jj}) = \chi_A(x).$$

The last equality is the diagonal-count result just proved. Thus the intrinsic definition agrees with the earlier two-by-two computational formula.

Definition (LADR 8.35)**.** A block-diagonal matrix $\operatorname{diag}(A_1, \ldots, A_r)$ has square matrices A_j along its diagonal and zeros outside these blocks. It represents an operator preserving the corresponding direct-sum coordinate subspaces.

Proposition (LADR 8.37)

A complex operator has a block-diagonal matrix with one upper-triangular block for each eigenvalue λ, of size d_λ, with every diagonal entry in that block equal to λ.

Proof. 1. By the nilpotence tests, each N_λ has a strictly upper-triangular matrix. Adding λI gives the required diagonal and block size $\dim G(\lambda, T) = d_\lambda$.
2. The generalized eigenspace direct sum makes the concatenation a basis. Invariance of each summand makes every off-block entry zero.

□

Example (LADR 8.24, 8.27, 8.38) 1. Over $\mathbb{C}$, let

$$A = \begin{pmatrix} 6 & 3 & 4 \\ 0 & 6 & 2 \\ 0 & 0 & 7 \end{pmatrix}, \quad A - 6I = \begin{pmatrix} 0 & 3 & 4 \\ 0 & 0 & 2 \\ 0 & 0 & 1 \end{pmatrix}, \quad (A - 6I)^2 = \begin{pmatrix} 0 & 0 & 10 \\ 0 & 0 & 2 \\ 0 & 0 & 1 \end{pmatrix}.$$

The first kernel requires $z = 0, y = 0$; the second requires only $z = 0$, as does the third power. Thus $E(6, A) = \mathbb{C}e_1$ and $G(6, A) = \operatorname{span}(e_1, e_2)$.

2. For 7, solve $(A - 7I)(x, y, z) = 0$: $-y + 2z = 0$, $-x + 3y + 4z = 0$ give $y = 2z, x = 10z$. Its generalized dimension is 1 because the direct sum has total dimension 3. Thus $G(7, A) = \mathbb{C}(10, 2, 1)$.
3. The dimensions give $\chi_A = (x - 6)^2(x - 7)$. There is no ordinary eigenbasis: the two ordinary eigenspaces have total dimension 2.
4. $Ae_1 = 6e_1$, $Ae_2 = 3e_1 + 6e_2$, $A(10, 2, 1) = 7(10, 2, 1)$. Hence in this basis the columns are

$$\begin{pmatrix} 6 & 3 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 7 \end{pmatrix}.$$

Example (LADR 8.36)

$$\left(\begin{array}{c|cc|cc} 4 & 0 & 0 & 0 & 0 \\ \hline 0 & 2 & -3 & 0 & 0 \\ 0 & 0 & 2 & 0 & 0 \\ \hline 0 & 0 & 0 & 1 & 7 \\ 0 & 0 & 0 & 0 & 1 \end{array}\right).$$

The three invariant coordinate groups have bases (e_1), (e_2, e_3), (e_4, e_5). The zero entries outside the blocks say that each basis vector maps into its own group. Counting diagonal entries gives $\chi_A = (x - 4)(x - 2)^2(x - 1)^2$.

Shortcuts

- For a triangular complex matrix, count diagonal occurrences to obtain χ_T; no determinant expansion is required.
- Algebraic multiplicity is the stabilized shifted-kernel dimension; geometric multiplicity uses only the first kernel.

Traps

- Algebraic and geometric multiplicities can differ: $\left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$ has algebraic multiplicity 2 but geometric multiplicity 1.
- Cayley–Hamilton does not make every vector an eigenvector: for $T = \operatorname{diag}(1, 2)$, $\chi_T(T)(1, 1) = 0$, while $T(1, 1) = (1, 2)$ is not a scalar multiple of $(1, 1)$.

§4 Jordan blocks and their sizes

The question. How do the shifted kernels specify a canonical matrix? A three-vector chain for $T - I$ gives $\left(\begin{smallmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{smallmatrix}\right)$, with successive kernel dimensions $1, 2, 3$.

Definition (Notes Definition 40; LADR 8.44)**.** The Jordan block $J_r(\lambda)$ is the r-by-r matrix with λ on its diagonal, 1 immediately above the diagonal and zeros elsewhere. A Jordan normal form is a block-diagonal matrix of such blocks; a basis giving that matrix is a Jordan basis. Different blocks may have the same eigenvalue. A block is $\lambda I + N$ with one length-r chain for N.

Proposition (Notes Proposition 76; LADR 8.46)

Every operator on a finite-dimensional complex space has a Jordan basis.

Proof. 1. On $G(\lambda, T)$, choose a chain basis for $N_\lambda = (T-\lambda I)|_G$ by Proposition 75. Its matrix consists of zero-diagonal unit-chain blocks.

2. Adding λI turns each into $J_r(\lambda)$. Concatenating bases over the direct sum in Proposition 73 gives a Jordan basis of V.

□

Example (Notes Chapter 15: displayed Jordan forms)

$$\left(\begin{array}{cc|ccc} 2 & 1 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 & 0 \\ \hline 0 & 0 & 3 & 1 & 0 \\ 0 & 0 & 0 & 3 & 1 \\ 0 & 0 & 0 & 0 & 3 \end{array}\right), \quad \left(\begin{array}{cc|cc|c} 2 & 1 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 & 0 \\ \hline 0 & 0 & 3 & 1 & 0 \\ 0 & 0 & 0 & 3 & 0 \\ \hline 0 & 0 & 0 & 0 & 3 \end{array}\right), \quad \begin{pmatrix} 2 & 0 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 & 0 \\ 0 & 0 & 3 & 0 & 0 \\ 0 & 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 0 & 3 \end{pmatrix}.$$

They have respectively two, three and five Jordan blocks. In the first, basis vectors 1–2 span $G(2, T)$, and 3–5 span $G(3, T)$; zero entries between those groups express invariance. In the second, $G(3, T)$ contains two separate chains. The last matrix has only length-1 chains and is diagonal.

Proposition (Jordan data: minimal polynomial and multiplicities)

For each eigenvalue λ, let its block lengths be $r_{\lambda,1}, \ldots, r_{\lambda,b_\lambda}$. Its geometric multiplicity is b_λ, its algebraic multiplicity is $\sum_j r_{\lambda,j}$, and its factor in m_T is $(x-\lambda)^{\max_j r_{\lambda,j}}$. Thus T is diagonalizable exactly when every block has size 1.

Proof. 1. A size-r block has $\ker(J_r(\lambda)-\lambda I) = Fe_1$ and its full r-dimensional space is generalized. On other-eigenvalue blocks the shift is invertible, since their triangular diagonal entries are nonzero. This proves both multiplicity formulas.

2. Write $p = (x-\lambda)^s q$ with $q(\lambda) \neq 0$. Division by $x-\lambda$ gives $q(x) = q(\lambda) + (x-\lambda)R(x)$. Substitute $\lambda I + N$ to get $q(\lambda I + N) = q(\lambda)I + NR(\lambda I + N)$.

3. This last operator is invertible: after factoring out $q(\lambda)$, it is $I + K$, where $K = q(\lambda)^{-1}NR(\lambda I + N)$ has $K^r = 0$ because its commuting factors include N^r; $(I+K)\sum_{j=0}^{r-1}(-K)^j = I - (-K)^r = I$. Thus $p(J) = N^s q(J)$ is zero precisely when $s \geq r$.

4. A polynomial annihilates a block-diagonal matrix exactly when it annihilates every block. The least monic product with the required multiplicity for each root has the stated maximal exponents. For each eigenvalue, geometric equals algebraic multiplicity exactly when the number of blocks equals their total size, hence exactly when every length is 1. Apply the multiplicity criterion proved above.

□

Proposition (Notes Proposition 77)

Suppose T admits a Jordan basis over F (in particular, this holds over $\mathbb{C}$). For integers $k \geq 1$, put $d_k = \dim \ker(T-\lambda I)^k$, with $d_0 = 0$. The number of blocks for λ of size at least k is $d_k - d_{k-1}$; the number of size exactly k is $2d_k - d_{k-1} - d_{k+1}$. Consequently Jordan form is unique up to block order.

Count what each extra power adds

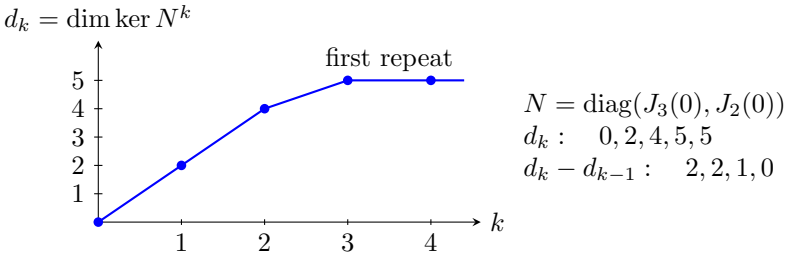
One chain. On $e_3 \mapsto e_2 \mapsto e_1 \mapsto 0$, $\ker N$ contains e_1; $\ker N^2$ adds e_2; $\ker N^3$ adds e_3. Later powers add nothing. A length-r chain therefore contributes one new direction at step k exactly when $r \geq k$.

Several chains. Add these contributions. The increase $d_k - d_{k-1}$ counts the chains still contributing at step k. Subtract the next increase to count the chains that end at this step, giving the number of length exactly k.

Proof. 1. On a block for λ, the shift sends e_j to e_{j-1} and the first $\min(k, r)$ vectors to zero after k steps. Thus its nullity is $\min(k, r)$. Other-eigenvalue blocks have invertible shift and contribute zero.

2. Subtracting the nullities at k and $k-1$ contributes 1 for each block with $r \geq k$, and 0 for each smaller block. Summing proves the first formula.
3. Subtract the counts for sizes at least $k+1$ from sizes at least k: $(d_k - d_{k-1}) - (d_{k+1} - d_k) = 2d_k - d_{k-1} - d_{k+1}$.
4. These numbers determine every block multiplicity for every eigenvalue, proving uniqueness up to order.

□



Each block adds one kernel direction per step until its length is reached. The first repeat prevents any later increase by Proposition 69.

Construct an actual Jordan basis (Notes Example 62)

1. Over $\mathbb{C}$, for $A = \left(\begin{smallmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{smallmatrix}\right)$ set $N = A - I$. Then $N^2 = \left(\begin{smallmatrix} 0 & 0 & 4 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{smallmatrix}\right)$ and $N^3 = 0$. The successive nullities are $1, 2, 3$, so one length-3 block occurs.
2. Choose $v_3 = e_3$ outside $\ker N^2 = \mathrm{span}(e_1, e_2)$ so $N^2 v_3 \neq 0$; chain independence then guarantees three basis vectors. Then $v_2 = Nv_3 = (3, 2, 0)$ and $v_1 = Nv_2 = (4, 0, 0)$.
3. With $S = (v_1\ v_2\ v_3) = \left(\begin{smallmatrix} 4 & 3 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{smallmatrix}\right)$, the equations $Av_1 = v_1$, $Av_2 = v_2 + v_1$, $Av_3 = v_3 + v_2$ give the three columns of $AS = SJ_3(1)$. Hence $A = SJ_3(1)S^{-1}$ without solving for the inverse.

$$v_3 = (0,0,1) \xrightarrow{N = A - I} v_2 = (3,2,0) \xrightarrow{N} v_1 = (4,0,0) \xrightarrow{N} 0$$

ordinary eigenvector

Starting at v_3, apply $A - I$ until the last nonzero vector v_1. Order the basis (v_1, v_2, v_3) to place the unit coefficients above the diagonal.

Recover sizes from nullities (Notes Example 63)

1. On $\mathbb{C}^8$, the shifted nullities for eigenvalue 1 are $d_0, d_1, d_2, d_3 = 0, 2, 4, 4$. The differences are $2, 2, 0$, so both blocks have length 2.
2. For eigenvalue 2 they start $0, 2, 3, 4$. These account for dimension 4, since the first generalized eigenspace already has dimension 4. The next nullity is therefore 4, and differences are $2, 1, 1, 0$. The sizes are 3 and 1.
3. Thus Jordan form is $\mathrm{diag}(J_2(1), J_2(1), J_3(2), J_1(2))$. The characteristic polynomial is $(x-1)^4(x-2)^4$ and the minimal polynomial is $(x-1)^2(x-2)^3$.

Example (Notes Chapter 15: generalized bases do not determine similarity)

Work over $\mathbb{C}$.

1. For $A = \left(\begin{smallmatrix} 3 & -1 & -1 \\ 2 & 0 & -1 \\ 2 & -1 & 0 \end{smallmatrix}\right)$, $N = A - I = \left(\begin{smallmatrix} 2 & -1 & -1 \\ 2 & -1 & -1 \\ 2 & -1 & -1 \end{smallmatrix}\right)$ has identical nonzero rows, so rank 1; multiplying gives $N^2 = 0$. Thus its blocks have lengths 2 and 1.
2. For $B = \left(\begin{smallmatrix} 3 & -2 & 0 \\ 1 & 1 & -1 \\ 0 & 2 & -1 \end{smallmatrix}\right)$, $M = B - I = \left(\begin{smallmatrix} 2 & -2 & 0 \\ 1 & 0 & -1 \\ 0 & 2 & -2 \end{smallmatrix}\right)$ has row 3 equal to 2 row 2 − row 1, while rows 1 and 2 are independent, so rank 2. Also $M^2 = \left(\begin{smallmatrix} 2 & -4 & 2 \\ 2 & -4 & 2 \\ 2 & -4 & 2 \end{smallmatrix}\right) \neq 0$ and $M^3 = 0$, so its sole block has length 3.
3. For both matrices every standard basis vector is a generalized eigenvector for 1, as it is for I_3. Nevertheless their shifted ranks are respectively 1, 2 and 0. Similarity preserves rank because an invertible change of coordinates preserves range dimension (Week 4), so these three matrices are pairwise nonsimilar.

Shortcuts

- Kernel first differences must be nonnegative and nonincreasing: they count blocks of size at least k. This quickly detects impossible data.
- The first nullity counts blocks, the final nullity gives their total size, and the last increase locates the largest size.
- To construct a longest chain, choose a vector in $\ker(T-\lambda I)^r$ outside the preceding kernel and apply the shift repeatedly.

Traps

- Algebraic multiplicity need not equal the exponent in the minimal polynomial: two blocks $J_2(1), J_2(1)$ give algebraic multiplicity 4 but exponent 2.
- A generalized eigenvector basis need not be a Jordan basis: the standard basis for the matrix in Example 62 consists of generalized eigenvectors, but its nonunit entries do not have Jordan form.

§5 Powers and square roots

The question. How do Jordan blocks make repeated application explicit? For $J = \left(\begin{smallmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{smallmatrix}\right)$, $J^2 = \left(\begin{smallmatrix} 1 & 2 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{smallmatrix}\right)$; the second superdiagonal records a quadratic factor in the exponent.

Proposition (Notes Chapter 15: powers and Jordan–Chevalley decomposition)

For $J_r(\lambda) = \lambda I + N$ and integer $k \geq 0$, $J_r(\lambda)^k = \sum_{j=0}^{\min(k,r-1)} \binom{k}{j} \lambda^{k-j} N^j$, with the convention $\lambda^0 = 1$. A Jordan matrix is $D+N$ with diagonal D, nilpotent N and $DN = ND$. If invertible it also equals $D(I + D^{-1}N)$, with commuting factors.

Proof. 1. For commuting A, B, induction on k gives $(A+B)^k = \sum_{j=0}^{k} \binom{k}{j} A^{k-j} B^j$: multiply by $A+B$, commute the factors, and combine adjacent terms using Pascal's identity.

2. Take $A = \lambda I$, $B = N$ and use $N^j = 0$ for $j \geq r$. Restricting to $j \leq k$ avoids negative exponents, including when $\lambda = 0$.
3. For a full Jordan matrix, let D equal its diagonal and $N = J - D$. On each block D is a scalar matrix, so it commutes with N; powers of the block shifts show N nilpotent.
4. If J is invertible, none of its diagonal eigenvalues is zero, so D is invertible and $D(I + D^{-1}N) = D + N$. Both factors commute because D and N do. These are the additive and multiplicative decompositions described in the notes.

□

Compute the hundredth power (Notes Chapter 15)

1. For $J = J_3(1)$, $N = J - I$ has $N^3 = 0$, so $J^{100} = I + 100N + \binom{100}{2} N^2$.
2. Since $\binom{100}{2} = 100 \cdot 99/2 = 4950$, $J^{100} = \left(\begin{smallmatrix} 1 & 100 & 4950 \\ 0 & 1 & 100 \\ 0 & 0 & 1 \end{smallmatrix}\right)$.

Definition (LADR 8C: square root)**.** An operator R is a square root of T if $R^2 = T$.

Proposition (LADR 8.39)

Over $\mathbb{R}$ or $\mathbb{C}$, if N is nilpotent, then $I + N$ has a square root which is a polynomial in N.

Why the coefficients can be chosen one at a time

Small case. For $N^3 = 0$, try $R = I + aN + bN^2$. Its square is

$$R^2 = I + 2aN + (a^2 + 2b)N^2.$$

Matching $I + N$ gives $2a = 1$, then $a^2 + 2b = 0$. Thus $a = 1/2$ and $b = -1/8$.

Choice. Add the next power only after fixing the earlier coefficients. The new $a_k N^k$ contributes $2a_k$ to degree k and cannot change any lower degree. Since $2 \neq 0$ in $\mathbb{R}$ or $\mathbb{C}$, its coefficient can always cancel the known degree-k error. Nilpotence makes this a finite calculation.

Proof. 1. Choose r with $N^r = 0$. If $r = 1$, use $R = I$. Otherwise seek $R = I + \sum_{j=1}^{r-1} a_j N^j$.

2. The coefficient of N in R^2 is $2a_1$, so set $a_1 = 1/2$. For $2 \leq k < r$, set $a_k = -\frac{1}{2} \sum_{j=1}^{k-1} a_j a_{k-j}$.

3. Then the coefficient of N^k in R^2 is $2a_k + \sum_{j=1}^{k-1} a_j a_{k-j} = 0$. All terms of degree at least r vanish. Hence $R^2 = I + N$. The first coefficients are $1/2, -1/8, 1/16$.

□

Proposition (LADR 8.41)

Every invertible operator on a finite-dimensional complex vector space has a square root.

Proof. 1. On $G(\lambda, T)$ write $T = \lambda(I + N_\lambda/\lambda)$. The scaled operator N_λ/λ is nilpotent, so the preceding result gives $S_\lambda^2 = I + N_\lambda/\lambda$.

2. Choose $c_\lambda \in \mathbb{C}$ with $c_\lambda^2 = \lambda$. Set $R_\lambda = c_\lambda S_\lambda$, so $R_\lambda^2 = T|_{G(\lambda,T)}$.

3. Define $R(\sum_\lambda v_\lambda) = \sum_\lambda R_\lambda v_\lambda$ using Proposition 73's unique decomposition. This is linear and $R^2 v = \sum_\lambda T v_\lambda = Tv$.

□

Every positive integer root (after LADR 8.41). For each integer $k \geq 1$, an invertible complex operator has a kth root.

1. For $N^r = 0$, seek $Q = I + \sum_{j=1}^{r-1} a_j N^j$ with $Q^k = I + N$. In the coefficient of N^j, the new a_j occurs exactly k times (choose it from one of the k factors); all other contributions use earlier coefficients. For $r > 1$, set $a_1 = 1/k$ and choose each later a_j to cancel those contributions, dividing by $k \neq 0$ in $\mathbb{C}$.
2. On $G(\lambda, T)$ factor $T = \lambda(I + N_\lambda/\lambda)$ and choose $c_\lambda^k = \lambda$ by the fundamental theorem of algebra. The operator $c_\lambda Q$ has kth power equal to this restriction. Combine these operators on the direct sum, exactly as in the square-root proof.

Construct a square root

1. For $J_3(1) = I + N$, $N^3 = 0$, so $R = I + N/2 - N^2/8 = \begin{pmatrix} 1 & 1/2 & -1/8 \\ 0 & 1 & 1/2 \\ 0 & 0 & 1 \end{pmatrix}$.
2. Expanding gives $R^2 = I + N + (1/4 - 1/4)N^2 = J_3(1)$, since all higher powers vanish.

Shortcuts

- Compute a power block by block, using only powers of each nilpotent shift up to its block size minus 1.
- A square root of $I + N$ is found by matching coefficients; no infinite-series convergence is needed because the operator expression is finite.

Traps

- An invertible real operator need not have a real square root: multiplication by -1 on $\mathbb{R}$ would require a real scalar r with $r^2 = -1$.
- A complex noninvertible operator may have no square root: if $R^2 = J_3(0)$, then $R^6 = 0$, so the dimension bound gives $R^3 = 0$. This forces $J_3(0)^2 = R^4 = 0$, contradicting its nonzero second power.

§6 Problem set

The question. Recover blocks from shifted kernels and use them for powers and minimal polynomials. For example, nullities $0, 1, 2, 3, 3$ give one block of length 3.

HW8 P1. For differentiation $D : \mathcal{P}_n(\mathbb{C}) \to \mathcal{P}_n(\mathbb{C})$, (a) prove nilpotence and determine its index; (b) determine the geometric and algebraic multiplicities of 0; (c) find a basis giving a single Jordan block.
Hint: Use the derivative chain $1, x, x^2/2!, \ldots$ and the nilpotence tests.

HW8 P2. Over $\mathbb{R}$ or $\mathbb{C}$, construct explicit matrices with (a) eigenvalue 2 of algebraic multiplicity 3 and geometric multiplicity 1, and eigenvalue -1 of algebraic and geometric multiplicity 2; (b) the same algebraic multiplicities but geometric multiplicity of 2 equal to 2. Give each minimal polynomial.
Hint: Use the Jordan-data formulas for number and largest size of blocks.

HW8 P3. Over F, for $J = \left(\begin{smallmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{smallmatrix}\right)$, (a) compute $\ker(J - I)$, $\ker(J - I)^2$, $\ker(J - I)^3$; (b) determine $G(1, J)$ and both multiplicities; (c) compute J^m for each positive integer m.
Hint: Use the chain basis and the commuting binomial formula.

HW8 P4. A nilpotent operator has one block of size 3 and one of size 2. (a) Compute $\dim \ker N^k$ for $k = 0, 1, 2, 3, 4$. (b) On the same five-dimensional space, if instead given only $\dim \ker N = 2$, $\dim \ker N^2 = 4$, $\dim \ker N^3 = 5$, explain how these force the same two block sizes.
Hint: Apply Proposition 77's first differences to these five-dimensional data.

HW8 P5 (optional). On $\mathbb{C}^6$, T has eigenvalues 1 and 2 with shifted nullities $1, 2, 2$ at powers $1, 2, 3$ for 1, and $2, 3, 4, 4$ at powers $1, 2, 3, 4$ for 2. Determine its Jordan form up to block order.
Hint: Stabilization and Proposition 77 recover lengths from successive differences.

HW8 P6 (optional). Over $\mathbb{C}$, for blocks $J_3(4), J_2(4), J_1(-1)$, determine the characteristic polynomial, minimal polynomial, both multiplicities of each eigenvalue, and diagonalizability.
Hint: Use sums of sizes for χ_T and maxima of sizes for m_T.

HW8 P7 (optional). Prove that for nilpotent N, $\dim \ker N^k - \dim \ker N^{k-1}$ counts the Jordan blocks of size at least k.
Hint: Use the single-block nullity $\min(k, r)$ from Proposition 77.

Exam practice (midterm2-review.pdf, Problem 5). For $A = \left(\begin{smallmatrix} 3 & 1 & 1 \\ 0 & 3 & -1 \\ 0 & 0 & 1 \end{smallmatrix}\right)$ as an operator $\mathbb{R}^3 \to \mathbb{R}^3$, find its minimal polynomial, characteristic polynomial, eigenvalues and eigenspaces. For the characteristic polynomial, regard the same real matrix on $\mathbb{C}^3$ and use the intrinsic definition above.
Hint: Use triangular diagonal counts, then solve shifted kernels and test the candidate annihilating factors.

Midterm 2 Review, Problem 8 (last two assertions). Decide true or false, with a proof or counterexample: (a) $\dim E(\lambda, T)$ equals the exponent of $x - \lambda$ in m_T; (b) for an upper-triangular matrix, the minimal polynomial depends on the entries above the diagonal (interpret as "can depend").
Hint: Compare the number of blocks with their largest length, and compare λI_2 with $J_2(\lambda)$.